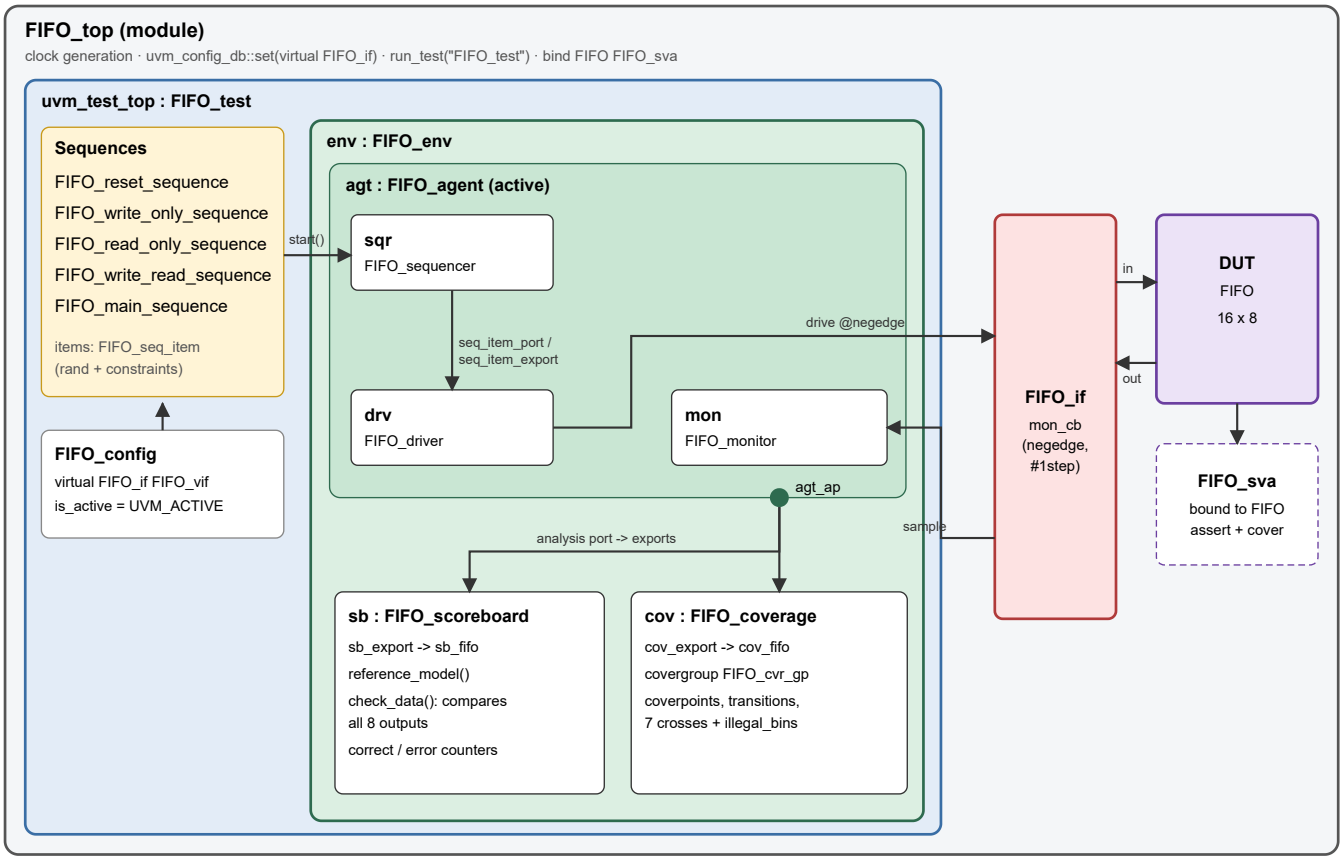
UVM Project
Ahmed Tarek Ahmed

Plan:

Label	Design Requirement Description	Stimulus Generation	Functional Coverage	Functionality Check
FIFO_1	Asynchronous reset: when rst_n = 0, wr_ptr, rd_ptr and count clear immediately (without waiting for a clock edge) and wr_ack, overflow and underflow are 0.	Directed: FIFO_reset_sequence at the start and again while the FIFO is not empty. Randomized: c_rst_n (rst_n = 0 with 5%) in FIFO_main_sequence.	Reset hit in empty, partially full and full states (FIFO_main_sequence).	Assertion (a_reset, deferred immediate). Scoreboard reference model.
FIFO_2	When wr_en is high and the FIFO is not full, data_in is written and wr_ack is asserted in the following cycle.	FIFO_write_only_sequence, FIFO_write_read_sequence, FIFO_main_sequence (c_wr_en: 70% / 30%).	cx_wr_ack (wr_en x rd_en x wr_ack)	Assertion (ap_wr_ack). Scoreboard.
FIFO_3	When no write is accepted (wr_en low or FIFO full) wr_ack is 0 in the following cycle.	FIFO_write_only_sequence (12 writes into 8 entries), FIFO_main_sequence.	cx_wr_ack	Assertion (ap_wr_ack_low). Scoreboard.
FIFO_4	A write attempted while the FIFO is full is ignored and overflow is asserted in the following cycle.	FIFO_write_only_sequence (4 writes while full), FIFO_main_sequence.	cx_overflow (wr_en x rd_en x overflow)	Assertion (ap_overflow). Scoreboard.
FIFO_5	overflow is 0 whenever the previous cycle had no rejected write (no sticky overflow).	FIFO_write_read_sequence / FIFO_read_only_sequence after the FIFO is full, FIFO_main_sequence.	cx_overflow	Assertion (ap_overflow_low). Scoreboard.
FIFO_6	A read attempted while the FIFO is empty is ignored and underflow is asserted in the following cycle (sequential output).	FIFO_read_only_sequence (12 reads from 8 entries), FIFO_main_sequence.	cx_underflow (wr_en x rd_en x underflow)	Assertion (ap_underflow). Scoreboard.
FIFO_7	underflow is 0 whenever the previous cycle had no rejected read.	FIFO_read_only_sequence, FIFO_main_sequence.	cx_underflow	Assertion (ap_underflow_low). Scoreboard.
FIFO_8	full is asserted if and only if count == FIFO_DEPTH.	FIFO_write_only_sequence, FIFO_main_sequence (write biased).	cx_full, cp_full_trans (0=>1, 1=>0)	Assertion (ap_full). Scoreboard.
FIFO_9	empty is asserted if and only if count == 0.	FIFO_reset_sequence, FIFO_read_only_sequence, FIFO_main_sequence (read biased).	cx_empty, cp_empty_trans	Assertion (ap_empty). Scoreboard.
FIFO_10	almostfull is asserted if and only if count == FIFO_DEPTH-1.	FIFO_write_only_sequence, FIFO_read_only_sequence, FIFO_main_sequence.	cx_almostfull, cp_almostfull_trans	Assertion (ap_almostfull). Scoreboard.
FIFO_11	almostempty is asserted if and only if count == 1.	FIFO_write_only_sequence, FIFO_read_only_sequence, FIFO_main_sequence.	cx_almostempty, cp_almostempty_trans	Assertion (ap_almostempty). Scoreboard.
FIFO_12	full and empty are never asserted together.	All sequences.	-	Assertion (ap_full_empty_mutex).
FIFO_13	Write only and not full: count + 1. Read only and not empty: count - 1. No request: count unchanged.	FIFO_write_only_sequence, FIFO_read_only_sequence, FIFO_main_sequence.	cx_full / cx_empty	Assertions (ap_count_wr, ap_count_rd, ap_count_idle).
FIFO_14	Simultaneous wr_en and rd_en while EMPTY: only the write takes place (count + 1, wr_ack = 1, underflow = 1).	FIFO_read_only_sequence (drain) followed by FIFO_write_read_sequence.	cx_underflow bin <1,1,1>, cx_wr_ack bin <1,1,1>	Assertion (ap_count_wr_rd_empty). Scoreboard.
FIFO_15	Simultaneous wr_en and rd_en while FULL: only the read takes place (count - 1, overflow = 1).	FIFO_write_only_sequence (fill) followed by FIFO_write_read_sequence.	cx_overflow bin <1,1,1>	Assertion (ap_count_wr_rd_full). Scoreboard.
FIFO_16	Simultaneous wr_en and rd_en while neither full nor empty: both take place, count unchanged.	FIFO_write_read_sequence (after the first item), FIFO_main_sequence.	cx_almostfull / cx_almostempty bins <1,1,x>	Assertion (ap_count_wr_rd_mid). Scoreboard.

FIFO_17	wr_ptr / rd_ptr increment on every accepted write / read, wrap to 0 after FIFO_DEPTH-1 and hold otherwise.	FIFO_write_only_sequence + FIFO_read_only_sequence (more than FIFO_DEPTH accesses), FIFO_main_sequence.	Cover directives of the pointer properties (sequential domain coverage)	Assertions (ap_wr_ptr_inc, ap_rd_ptr_inc, ap_wr_ptr_wrap, ap_rd_ptr_wrap, ap_wr_ptr_hold, ap_rd_ptr_hold).
FIFO_18	Pointers never exceed FIFO_DEPTH-1 and count never exceeds FIFO_DEPTH.	All sequences.	-	Assertions (ap_ptr_range, ap_count_range).
FIFO_19	Data integrity: data_out returns the data in the same order it was written (first in, first out).	All sequences; c_data_in biases data_in to 0x0000, 0xFFFF and random values.	cp_data_in (zero, all_ones, others) on accepted writes	Golden/reference model (FIFO_scoreboard.reference_m
FIFO_20	data_out holds its value when no valid read takes place.	FIFO_write_only_sequence, idle cycles of FIFO_main_sequence.	-	Assertion (ap_data_out_hold). Scoreboard.
FIFO_21	Every reachable combination of wr_en, rd_en and each output flag is exercised.	FIFO_main_sequence (write biased 2000 items + read biased 1000 items) + directed sequences.	7 crosses: cx_full, cx_empty, cx_almostfull, cx_almostempty, cx_overflow, cx_underflow, cx_wr_ack (impossible combinations are illegal_bins)	Functional coverage only
FIFO_22	Occupancy transitions empty -> almostempty -> almostfull -> full and back are exercised.	FIFO_write_only_sequence then FIFO_read_only_sequence, FIFO_main_sequence.	cp_full_trans, cp_empty_trans, cp_almostfull_trans, cp_almostempty_trans; cover directives cp_fill_to_full, cp_drain_to_empty	Functional coverage + flag assertions

UVM Testbench Structure:



The editable diagram is provided as **Docs/FIFO_UVM_structure.drawio** (draw.io).

How the UVM testbench works:

1. Top module (FIFO_top)

FIFO_top generates a 10 ns clock and instantiates the interface FIFO_if and the DUT FIFO. The DUT ports are connected to the interface signals. The assertion module FIFO_sva is attached with bind FIFO FIFO_sva ..., so the assertions see the DUT ports and the internal wr_ptr, rd_ptr and count without changing the RTL. In an initial block the top stores the interface in the configuration database (uvm_config_db#(virtual FIFO_if)::set(null, "uvm_test_top", "FIFO_IF", FIFOif)) and calls run_test("FIFO_test").

2. Test (FIFO_test)

In build_phase the test creates the environment and the five sequences, and gets the virtual interface from the config_db. It places it in a FIFO_config object, sets the agent to UVM_ACTIVE and shares the object with all components (set(this, "*", "CFG", FIFO_cfg)). In run_phase it raises an objection and starts the sequences on the agent sequencer in this order:

1. reset;
2. write_only, which fills the FIFO and then overflows it;
3. write_read while full;
4. read_only, which drains the FIFO and then underflows it;
5. write_read while empty;
6. reset in the middle of operation;

7. main_sequence, write biased (2000 items);
8. main_sequence, read biased (1000 items).

It then drops the objection, which ends the test.

3. Sequence item and sequences (stimulus generation)

FIFO_seq_item holds the randomized inputs (rst_n, wr_en, rd_en, data_in) and the observed outputs. It has four constraints:

- c_rst_n: rst_n is low 5% of the time.
- c_wr_en and c_rd_en: distributions controlled by the knobs WR_EN_ON_DIST and RD_EN_ON_DIST (default 70/30).
- c_data_in: biases data_in towards 0x0000 and 0xFFFF.

Each sequence creates items and sends them with start_item / finish_item:

- Directed sequences pin the control bits with inline constraints, for example randomize() with {rst_n==1; wr_en==1; rd_en==0;}.
- The main sequence uses the item constraints only.

4. Agent: sequencer, driver and monitor

FIFO_agent gets the configuration object and builds the sequencer and driver (active agent) and the monitor. In connect_phase it:

- connects drv.seq_item_port to sqr.seq_item_export;
- passes the virtual interface to the driver and the monitor;
- connects the monitor analysis port to the agent analysis port agt_ap.

Driving the interface: the driver calls get_next_item() and puts rst_n, wr_en, rd_en and data_in on the interface. It then waits for the next negative edge and calls item_done(). Inputs therefore change on the negative edge and are stable at the positive edge where the DUT samples them. Because rst_n is asynchronous, a reset item takes effect immediately.

Monitoring: the monitor waits on the clocking block mon_cb, which samples at the negative edge with #1step. One sample captures:

- the inputs driven at the previous negative edge;
- the outputs the DUT produced at the positive edge in between.

Sampling this way avoids any race with the driver. The monitor copies all 12 signals into a new FIFO_seq_item and broadcasts it on its analysis port.

5. Analysis: scoreboard and coverage collector

Scoreboard: FIFO_scoreboard receives the transactions through sb_export and a uvm_tlm_analysis_fifo. For each transaction, reference_model() updates a golden model written from the specification, not copied from the RTL. The model holds a memory array, the pointers and the count, and handles:

- asynchronous reset;
- a write when full, which is rejected and raises overflow;

- a read when empty, which is rejected and raises underflow;
- a simultaneous read and write when full or empty, where only one operation takes place.

From the new count it produces the expected data_out, wr_ack, overflow, underflow, full, empty, almostfull and almostempty. check_data() compares all eight outputs with !==. It increments correct_count or reports a uvm_error showing the expected and actual values. The totals are printed in report_phase.

Coverage collector: FIFO_coverage receives the same transactions through cov_export and samples the covergroup FIFO_cvr_gp whenever rst_n is high. The covergroup contains:

- coverpoints for every control input and output;
- transition bins on the four status flags;
- a data_in corner coverpoint;
- seven crosses of wr_en x rd_en x each output. Combinations the specification makes impossible are declared as illegal_bins, so reaching one is reported as an error.

6. Assertions

While the scoreboard checks the outputs cycle by cycle, FIFO_sva checks the timing relations and the internal state (pointers, counter, wrap-around, simultaneous access). The cover directives in the same module show that every property was exercised: this is the sequential domain coverage.

Assertions:

Feature	Assertion
Asynchronous reset clears wr_ptr, rd_ptr, count, wr_ack, overflow and underflow immediately	`always_comb if (!rst_n) assert final (wr_ptr==0 && rd_ptr==0 &&count==0 && ! wr_ack && !overflow && !underflow)`
A write while the FIFO is not full is acknowledged in the next cycle	`@(posedge clk) (wr_en && !full) |=> wr_ack`
Whenever a write is not accepted (e.g. the FIFO is full), wr_ack is 0	`@(posedge clk) !(wr_en && !full) |=> !wr_ack`
A write while the FIFO is full raises overflow	`@(posedge clk) (wr_en && full) |=> overflow`
overflow is low when no write was rejected	`@(posedge clk) !(wr_en && full) |=> !overflow`
A read while the FIFO is empty raises underflow	`@(posedge clk) (rd_en && empty) |=> underflow`
underflow is low when no read was rejected	`@(posedge clk) !(rd_en && empty) |=> !underflow`
full is high only when count == FIFO_DEPTH	`@(posedge clk) full == (count == FIFO_DEPTH)`
empty is high only when count == 0	`@(posedge clk) empty == (count == 0)`
almostfull is high only when count == FIFO_DEPTH-1	`@(posedge clk) almostfull == (count == FIFO_DEPTH-1)`
almostempty is high only when count == 1	`@(posedge clk) almostempty == (count == 1)`
full and empty are never high together	`@(posedge clk) !(full && empty)`
Write only (not full) increments count	`@(posedge clk) (wr_en && !rd_en && !full) |=> count == $past(count)+1`
Read only (not empty) decrements count	`@(posedge clk) (!wr_en && rd_en && !empty) |=> count == $past(count)-1`
No request keeps count	`@(posedge clk) (!wr_en && !rd_en) |=> $stable(count)`
Write + read while empty: only the write happens	`@(posedge clk) (wr_en && rd_en && empty) |=> count == $past(count)+1`
Write + read while full: only the read happens	`@(posedge clk) (wr_en && rd_en && full) |=> count == $past(count)-1`
Write + read otherwise: count unchanged	`@(posedge clk) (wr_en && rd_en && !full && !empty) |=> $stable(count)`
wr_ptr increments (mod FIFO_DEPTH) on an accepted write	`@(posedge clk) (wr_en && !full) |=> wr_ptr == ($past(wr_ptr)+1) % FIFO_DEPTH`
rd_ptr increments (mod FIFO_DEPTH) on an accepted read	`@(posedge clk) (rd_en && !empty) |=> rd_ptr == ($past(rd_ptr)+1) % FIFO_DEPTH`
wr_ptr wraps around to 0 after FIFO_DEPTH-1	`@(posedge clk) (wr_en && !full && wr_ptr == FIFO_DEPTH-1) |=> wr_ptr == 0`
rd_ptr wraps around to 0 after FIFO_DEPTH-1	`@(posedge clk) (rd_en && !empty && rd_ptr == FIFO_DEPTH-1) |=> rd_ptr == 0`
wr_ptr holds when no write is accepted	`@(posedge clk) !(wr_en && !full) |=> $stable(wr_ptr)`
rd_ptr holds when no read is accepted	`@(posedge clk) !(rd_en && !empty) |=> $stable(rd_ptr)`
Pointers stay inside the memory	`@(posedge clk) (wr_ptr < FIFO_DEPTH) && (rd_ptr < FIFO_DEPTH)`
count never exceeds FIFO_DEPTH	`@(posedge clk) count <= FIFO_DEPTH`
data_out holds when no read is accepted	`@(posedge clk) !(rd_en && !empty) |=> $stable(data_out)`

Bug Report:

Each bug was checked on its own: the full FIFO_test was run with only that bug in the DUT. The scoreboard checks 3035 transactions in each run. With every fix applied, the run passes with **0 errors**.

#	Bug	Detected by	Fix	Found in
1	almostfull was asserted at count == FIFO_DEPTH-2 instead of FIFO_DEPTH-1 (off by one).	SV project. Re-checked in UVM: scoreboard (506 failed transactions), ap_almostfull.	`assign almostfull = (count == FIFO_DEPTH-1);`	SV + UVM
2	underflow was a combinational assign (empty && rd_en) while the specification defines it as a sequential output.	SV project (already fixed there). Kept covered in UVM by ap_underflow, ap_underflow_low and the scoreboard.	`underflow moved into the read always block and registered on the clock edge.`	SV
3	wr_ack and overflow were not cleared by the asynchronous reset (they kept their old value / X after reset).	UVM: scoreboard (96 failed transactions, all during reset), a_reset.	`wr_ack <= 0; overflow <= 0; added to the reset branch of the write block.`	UVM (new)
4	overflow stayed stuck at 1 after a successful write because the write branch never cleared it.	UVM: scoreboard (96 failed transactions), ap_overflow_low.	`overflow <= 0; added to the accepted write branch.`	UVM (new)
5	Simultaneous wr_en and rd_en while FULL: the read took place but count was not decremented (count stuck at 8, data lost).	UVM: FIFO_write_read_sequence after FIFO_write_only_sequence, scoreboard (536 failed transactions), ap_count_wr_rd_full.	`else if ({wr_en, rd_en} == 2'b11 && full) count <= count - 1;`	UVM (new)
6	Simultaneous wr_en and rd_en while EMPTY: the write took place but count was not incremented (data written but FIFO still empty).	UVM: FIFO_write_read_sequence after FIFO_read_only_sequence, scoreboard (1493 failed transactions), ap_count_wr_rd_empty.	`else if ({wr_en, rd_en} == 2'b11 && empty) count <= count + 1;`	UVM (new)

Code Coverage:

```
Coverage Report by instance with details

================================================================================
=== Instance: /FIFO_top/DUT/FIFO_sva_inst
=== Design Unit: work.FIFO_sva
================================================================================

Assertion Coverage:
    Assertions                      27        27         0   100.00%
--------------------------------------------------------------------------------------------------------------------------------
Name                 File(Line)                   Failure      Pass     Vacuous     Disable     Attempt      Active Peak Active ATV
                                                  Count        Count    Count       Count       Count        Count  Count
--------------------------------------------------------------------------------------------------------------------------------
/FIFO_top/DUT/FIFO_sva_inst/a_reset
                     FIFO_sva.sv(21)                    0          144          0           0         144           0           0 off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ack
                     FIFO_sva.sv(73)                    0         1277       1540         219        3036           0           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ack_low
                     FIFO_sva.sv(75)                    0         1463       1344         228        3036           1           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_overflow
                     FIFO_sva.sv(77)                    0          263       2608         165        3036           0           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_overflow_low
                     FIFO_sva.sv(79)                    0         2477        276         282        3036           1           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_underflow
                     FIFO_sva.sv(81)                    0          415       2452         169        3036           0           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_underflow_low
                     FIFO_sva.sv(83)                    0         2325        432         278        3036           1           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_full
                     FIFO_sva.sv(85)                    0         2884          0         152        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_empty
                     FIFO_sva.sv(87)                    0         2884          0         152        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_almostfull
                     FIFO_sva.sv(89)                    0         2884          0         152        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_almostempty
                     FIFO_sva.sv(91)                    0         2884          0         152        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_full_empty_mutex
                     FIFO_sva.sv(93)                    0         2884          0         152        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_wr
                     FIFO_sva.sv(95)                    0          778       2065         193        3036           0           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_rd
                     FIFO_sva.sv(97)                    0          338       2531         166        3036           1           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_idle
                     FIFO_sva.sv(99)                    0          610       2235         191        3036           0           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_wr_rd_empty
                     FIFO_sva.sv(101)                   0          163       2714         159        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_wr_rd_full
                     FIFO_sva.sv(103)                   0           88       2794         154        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_wr_rd_mid
                     FIFO_sva.sv(105)                   0          336       2529         171        3036           0           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ptr_inc
                     FIFO_sva.sv(107)                   0         1277       1540         219        3036           0           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_rd_ptr_inc
                     FIFO_sva.sv(109)                   0          762       2086         187        3036           1           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ptr_wrap
                     FIFO_sva.sv(111)                   0          109       2770         157        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_rd_ptr_wrap
                     FIFO_sva.sv(113)                   0           53       2829         154        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ptr_hold
                     FIFO_sva.sv(115)                   0         1463       1344         228        3036           1           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_rd_ptr_hold
                     FIFO_sva.sv(117)                   0         1978        798         260        3036           0           2 off
/FIFO_top/DUT/FIFO_sva_inst/ap_ptr_range
                     FIFO_sva.sv(119)                   0         2884          0         152        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_range
                     FIFO_sva.sv(121)                   0         2884          0         152        3036           0           1 off
/FIFO_top/DUT/FIFO_sva_inst/ap_data_out_hold
                     FIFO_sva.sv(123)                   0         1978        798         260        3036           0           2 off

Directive Coverage:
    Directives                      27        27         0   100.00%
```

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DIRECTIVE COVERAGE:
--------------------------------------------------------------------------------------------
Name                                    Design Design   Lang File(Line)      Hits Status
                                        Unit   UnitType
--------------------------------------------------------------------------------------------
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ack    FIFO_sva Verilog  SVA  FIFO_sva.sv(129)1277 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ack_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(130)1463 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow  FIFO_sva Verilog  SVA  FIFO_sva.sv(131) 263 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(132)2477 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow FIFO_sva Verilog  SVA  FIFO_sva.sv(133) 415 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(134)2325 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_full      FIFO_sva Verilog  SVA  FIFO_sva.sv(135) 399 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_empty     FIFO_sva Verilog  SVA  FIFO_sva.sv(136) 755 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_almostfull
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(137) 309 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_almostempty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(138) 529 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr  FIFO_sva Verilog  SVA  FIFO_sva.sv(139) 778 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_rd  FIFO_sva Verilog  SVA  FIFO_sva.sv(140) 338 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_idle
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(141) 610 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_empty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(142) 163 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_full
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(143)  88 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_mid
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(144) 336 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_inc
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(145)1277 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_inc
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(146) 762 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_wrap
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(147) 109 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_wrap
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(148)  53 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(149)1463 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(150)1978 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_data_out_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(151)1978 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_fill_to_full
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(153)  64 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_drain_to_empty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(154)   5 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow_burst
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(155) 121 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow_burst
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(156) 170 Covered

================================================================================
=== Instance: /FIFO_top/DUT
=== Design Unit: work.FIFO
================================================================================
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        25        25         0   100.00%

---------------------------------Branch Details---------------------------------
```

ASSERTION RESULTS:

Name	File(Line)	Failure Count	Pass Count	Vacuous Count	Disable Count	Attempt Count	Active Count	Peak Active Count	ATV
/FIFO_top/DUT/FIFO_sva_inst/a_reset	FIFO_sva.sv(21)	0	144	0	0	144	0	0	off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ack	FIFO_sva.sv(73)	0	1277	1540	219	3036	0	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ack_low	FIFO_sva.sv(75)	0	1463	1344	228	3036	1	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_overflow	FIFO_sva.sv(77)	0	263	2608	165	3036	0	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_overflow_low	FIFO_sva.sv(79)	0	2477	276	282	3036	1	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_underflow	FIFO_sva.sv(81)	0	415	2452	169	3036	0	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_underflow_low	FIFO_sva.sv(83)	0	2325	432	278	3036	1	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_full	FIFO_sva.sv(85)	0	2884	0	152	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_empty	FIFO_sva.sv(87)	0	2884	0	152	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_almostfull	FIFO_sva.sv(89)	0	2884	0	152	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_almostempty	FIFO_sva.sv(91)	0	2884	0	152	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_full_empty_mutex	FIFO_sva.sv(93)	0	2884	0	152	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_wr	FIFO_sva.sv(95)	0	778	2065	193	3036	0	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_rd	FIFO_sva.sv(97)	0	338	2531	166	3036	1	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_idle	FIFO_sva.sv(99)	0	610	2235	191	3036	0	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_wr_rd_empty	FIFO_sva.sv(101)	0	163	2714	159	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_wr_rd_full	FIFO_sva.sv(103)	0	88	2794	154	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_wr_rd_mid	FIFO_sva.sv(105)	0	336	2529	171	3036	0	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ptr_inc	FIFO_sva.sv(107)	0	1277	1540	219	3036	0	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_rd_ptr_inc	FIFO_sva.sv(109)	0	762	2086	187	3036	1	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ptr_wrap	FIFO_sva.sv(111)	0	109	2770	157	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_rd_ptr_wrap	FIFO_sva.sv(113)	0	53	2829	154	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_wr_ptr_hold	FIFO_sva.sv(115)	0	1463	1344	228	3036	1	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_rd_ptr_hold	FIFO_sva.sv(117)	0	1978	798	260	3036	0	2	off
/FIFO_top/DUT/FIFO_sva_inst/ap_ptr_range	FIFO_sva.sv(119)	0	2884	0	152	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_count_range	FIFO_sva.sv(121)	0	2884	0	152	3036	0	1	off
/FIFO_top/DUT/FIFO_sva_inst/ap_data_out_hold	FIFO_sva.sv(123)	0	1978	798	260	3036	0	2	off

Total Coverage By Instance (filtered view): 100.00%

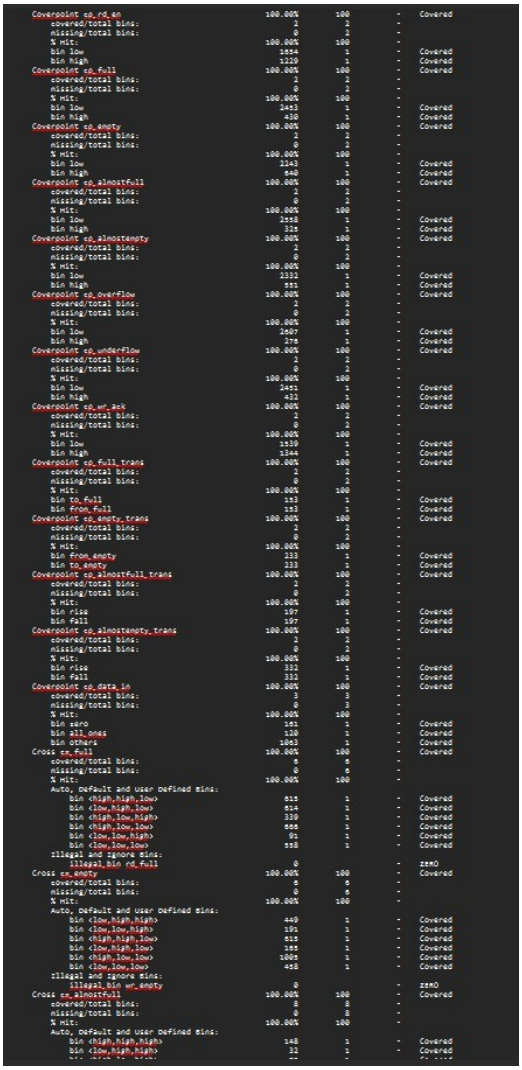
Functional Coverage:



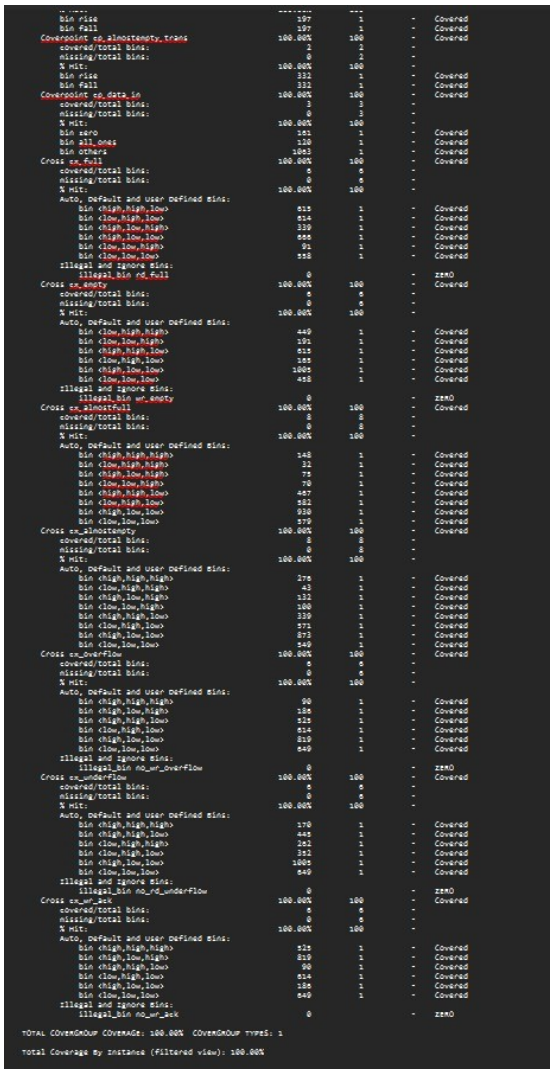
```
Coverage Report by instance with details

=================================================================================
=== Instance: /FIFO_coverage_pkg
=== Design Unit: work.FIFO_coverage_pkg
=================================================================================

Covergroup Coverage:
    Covergroups                 1        na        na   100.00%
        Coverpoints/Crosses    21        na        na        na
            Covergroup bins    75        75         0   100.00%
---------------------------------------------------------------------------------------------------------------
Covergroup                                          Metric       Goal       Bins    Status
---------------------------------------------------------------------------------------------------------------
 TYPE /FIFO_coverage_pkg/FIFO_coverage/FIFO_cvr_gp  100.00%        100          -    Covered
  covered/total bins:                                    75         75          -
  missing/total bins:                                     0         75          -
  % Hit:                                            100.00%        100          -
  Coverpoint cp_wr_en                               100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_rd_en                               100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_full                                100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_empty                               100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_almostfull                          100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_almostempty                         100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_overflow                            100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_underflow                           100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_wr_ack                              100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_full_trans                          100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_empty_trans                         100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_almostfull_trans                    100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_almostempty_trans                   100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
      missing/total bins:                                 0          1          -
      % Hit:                                        100.00%        100          -
  Coverpoint cp_data_in                             100.00%        100          -    Covered
      covered/total bins:                                 3          3          -
      missing/total bins:                                 0          3          -
      % Hit:                                        100.00%        100          -
  Cross cx_full                                     100.00%        100          -    Covered
      covered/total bins:                                 8          8          -
      missing/total bins:                                 0          8          -
      % Hit:                                        100.00%        100          -
  Cross cx_empty                                    100.00%        100          -    Covered
      covered/total bins:                                 8          8          -
      missing/total bins:                                 0          8          -
      % Hit:                                        100.00%        100          -
  Cross cx_almostfull                               100.00%        100          -    Covered
      covered/total bins:                                 8          8          -
      missing/total bins:                                 0          8          -
      % Hit:                                        100.00%        100          -
  Cross cx_almostempty                              100.00%        100          -    Covered
      covered/total bins:                                 8          8          -
      missing/total bins:                                 0          8          -
      % Hit:                                        100.00%        100          -
  Cross cx_overflow                                 100.00%        100          -    Covered
      covered/total bins:                                 8          8          -
      missing/total bins:                                 0          8          -
      % Hit:                                        100.00%        100          -
  Cross cx_underflow                                100.00%        100          -    Covered
      covered/total bins:                                 8          8          -
      missing/total bins:                                 0          8          -
      % Hit:                                        100.00%        100          -
  Cross cx_wr_ack                                   100.00%        100          -    Covered
      covered/total bins:                                 8          8          -
      missing/total bins:                                 0          8          -
      % Hit:                                        100.00%        100          -
Covergroup instance \/FIFO_coverage_pkg::FIFO_coverage::FIFO_cvr_gp
                                                    100.00%        100          -    Covered
  covered/total bins:                                    75         75          -
  missing/total bins:                                     0         75          -
  % Hit:                                            100.00%        100          -
  Coverpoint cp_wr_en                               100.00%        100          -    Covered
      covered/total bins:                                 1          1          -
```



```
Coverpoint cp_rd_en                               100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin low                                          1854          1          -    Covered
    bin high                                         1220          1          -    Covered
Coverpoint cp_full                                100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin low                                          2452          1          -    Covered
    bin high                                          430          1          -    Covered
Coverpoint cp_empty                               100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin low                                          2243          1          -    Covered
    bin high                                          640          1          -    Covered
Coverpoint cp_almostfull                          100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin low                                          2558          1          -    Covered
    bin high                                          325          1          -    Covered
Coverpoint cp_almostempty                         100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin low                                          2331          1          -    Covered
    bin high                                          551          1          -    Covered
Coverpoint cp_overflow                            100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin low                                          2607          1          -    Covered
    bin high                                          276          1          -    Covered
Coverpoint cp_underflow                           100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin low                                          2451          1          -    Covered
    bin high                                          432          1          -    Covered
Coverpoint cp_wr_ack                              100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin low                                          2139          1          -    Covered
    bin high                                         1344          1          -    Covered
Coverpoint cp_full_trans                          100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin to_full                                       153          1          -    Covered
    bin from_full                                     153          1          -    Covered
Coverpoint cp_empty_trans                         100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin from_empty                                    233          1          -    Covered
    bin to_empty                                      233          1          -    Covered
Coverpoint cp_almostfull_trans                    100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin rise                                          197          1          -    Covered
    bin fall                                          197          1          -    Covered
Coverpoint cp_almostempty_trans                   100.00%        100          -    Covered
    covered/total bins:                                   1          1          -
    missing/total bins:                                   0          1          -
    % Hit:                                          100.00%        100          -
    bin rise                                          331          1          -    Covered
    bin fall                                          331          1          -    Covered
Coverpoint cp_data_in                             100.00%        100          -    Covered
    covered/total bins:                                   3          3          -
    missing/total bins:                                   0          3          -
    % Hit:                                          100.00%        100          -
    bin zero                                          161          1          -    Covered
    bin all_ones                                      128          1          -    Covered
    bin others                                       1063          1          -    Covered
Cross cx_full                                     100.00%        100          -    Covered
    covered/total bins:                                   8          8          -
    missing/total bins:                                   0          8          -
    % Hit:                                          100.00%        100          -
    Auto, Default and User Defined bins:
        bin <high,high,low>                           615          1          -    Covered
        bin <low,high,low>                            614          1          -    Covered
        bin <high,low,high>                           339          1          -    Covered
        bin <high,low,low>                            666          1          -    Covered
        bin <low,low,high>                             91          1          -    Covered
        bin <low,low,low>                             558          1          -    Covered
    Illegal and Ignore bins:
        Illegal_bin rd_full                             0          -          -    ZERO
Cross cx_empty                                    100.00%        100          -    Covered
    covered/total bins:                                   8          8          -
    missing/total bins:                                   0          8          -
    % Hit:                                          100.00%        100          -
    Auto, Default and User Defined bins:
        bin <low,high,high>                           449          1          -    Covered
        bin <low,low,high>                            191          1          -    Covered
        bin <high,high,low>                           615          1          -    Covered
        bin <low,high,low>                            185          1          -    Covered
        bin <high,low,low>                           1005          1          -    Covered
        bin <low,low,low>                             458          1          -    Covered
    Illegal and Ignore bins:
        Illegal_bin wr_empty                            0          -          -    ZERO
Cross cx_almostfull                               100.00%        100          -    Covered
    covered/total bins:                                   8          8          -
    missing/total bins:                                   0          8          -
    % Hit:                                          100.00%        100          -
    Auto, Default and User Defined bins:
        bin <high,high,high>                          148          1          -    Covered
        bin <low,high,high>                            32          1          -    Covered
```



```
        bin rise                                      197          1          -    Covered
        bin fall                                      197          1          -    Covered
    Coverpoint cp_almostempty_trans               100.00%        100          -    Covered
        covered/total bins:                             2          2          -
        missing/total bins:                             0          2          -
        % Hit:                                    100.00%        100          -
        bin rise                                      331          1          -    Covered
        bin fall                                      331          1          -    Covered
    Coverpoint cp_data_in                         100.00%        100          -    Covered
        covered/total bins:                             3          3          -
        missing/total bins:                             0          3          -
        % Hit:                                    100.00%        100          -
        bin zero                                      161          1          -    Covered
        bin all_ones                                  128          1          -    Covered
        bin others                                   1063          1          -    Covered
    Cross cx_full                                 100.00%        100          -    Covered
        covered/total bins:                             8          8          -
        missing/total bins:                             0          8          -
        % Hit:                                    100.00%        100          -
        Auto, Default and User Defined bins:
            bin <high,high,low>                       615          1          -    Covered
            bin <low,high,low>                        614          1          -    Covered
            bin <high,low,high>                       339          1          -    Covered
            bin <high,low,low>                        666          1          -    Covered
            bin <low,low,high>                         91          1          -    Covered
            bin <low,low,low>                         558          1          -    Covered
        Illegal and Ignore bins:
            Illegal_bin rd_full                         0          -          -    ZERO
    Cross cx_empty                                100.00%        100          -    Covered
        covered/total bins:                             8          8          -
        missing/total bins:                             0          8          -
        % Hit:                                    100.00%        100          -
        Auto, Default and User Defined bins:
            bin <low,high,high>                       449          1          -    Covered
            bin <low,low,high>                        191          1          -    Covered
            bin <high,high,low>                       615          1          -    Covered
            bin <low,high,low>                        185          1          -    Covered
            bin <high,low,low>                       1005          1          -    Covered
            bin <low,low,low>                         458          1          -    Covered
        Illegal and Ignore bins:
            Illegal_bin wr_empty                        0          -          -    ZERO
    Cross cx_almostfull                           100.00%        100          -    Covered
        covered/total bins:                             8          8          -
        missing/total bins:                             0          8          -
        % Hit:                                    100.00%        100          -
        Auto, Default and User Defined bins:
            bin <high,high,high>                      148          1          -    Covered
            bin <low,high,high>                        32          1          -    Covered
            bin <high,high,low>                       615          1          -    Covered
            bin <low,high,low>                         75          1          -    Covered
            bin <low,low,high>                         78          1          -    Covered
            bin <high,low,low>                        467          1          -    Covered
            bin <low,low,low>                         582          1          -    Covered
            bin <high,low,low>                        930          1          -    Covered
            bin <low,low,low>                         579          1          -    Covered
    Cross cx_almostempty                          100.00%        100          -    Covered
        covered/total bins:                             8          8          -
        missing/total bins:                             0          8          -
        % Hit:                                    100.00%        100          -
        Auto, Default and User Defined bins:
            bin <high,high,high>                      378          1          -    Covered
            bin <low,high,high>                        43          1          -    Covered
            bin <high,low,high>                       131          1          -    Covered
            bin <low,low,high>                        100          1          -    Covered
            bin <high,high,low>                       339          1          -    Covered
            bin <low,high,low>                        571          1          -    Covered
            bin <high,low,low>                        573          1          -    Covered
            bin <low,low,low>                         549          1          -    Covered
    Cross cx_overflow                             100.00%        100          -    Covered
        covered/total bins:                             8          8          -
        missing/total bins:                             0          8          -
        % Hit:                                    100.00%        100          -
        Auto, Default and User Defined bins:
            bin <high,high,high>                       90          1          -    Covered
            bin <high,low,high>                       185          1          -    Covered
            bin <high,high,low>                       525          1          -    Covered
            bin <low,high,low>                        614          1          -    Covered
            bin <high,low,low>                        819          1          -    Covered
            bin <low,low,low>                         549          1          -    Covered
        Illegal and Ignore bins:
            Illegal_bin no_wr_overflow                  0          -          -    ZERO
    Cross cx_underflow                            100.00%        100          -    Covered
        covered/total bins:                             8          8          -
        missing/total bins:                             0          8          -
        % Hit:                                    100.00%        100          -
        Auto, Default and User Defined bins:
            bin <high,high,high>                      170          1          -    Covered
            bin <high,high,low>                       445          1          -    Covered
            bin <low,high,high>                       262          1          -    Covered
            bin <low,high,low>                        352          1          -    Covered
            bin <high,low,low>                       1005          1          -    Covered
            bin <low,low,low>                         549          1          -    Covered
        Illegal and Ignore bins:
            Illegal_bin no_rd_underflow                 0          -          -    ZERO
    Cross cx_wr_ack                               100.00%        100          -    Covered
        covered/total bins:                             8          8          -
        missing/total bins:                             0          8          -
        % Hit:                                    100.00%        100          -
        Auto, Default and User Defined bins:
            bin <high,high,high>                      525          1          -    Covered
            bin <high,low,high>                       819          1          -    Covered
            bin <high,high,low>                        90          1          -    Covered
            bin <low,high,low>                        614          1          -    Covered
            bin <high,low,low>                        186          1          -    Covered
            bin <low,low,low>                         549          1          -    Covered
        Illegal and Ignore bins:
            Illegal_bin no_wr_ack                       0          -          -    ZERO

TOTAL COVERGROUP COVERAGE: 100.00%  COVERGROUP TYPES: 1

Total Coverage By Instance (filtered view): 100.00%
```

Sequential Domain Coverage (cover directives):

```
Coverage Report by instance with details

================================================================================
=== Instance: /FIFO_top/DUT/FIFO_sva_inst
=== Design Unit: work.FIFO_sva
================================================================================

Directive Coverage:
    Directives                      27        27         0   100.00%

DIRECTIVE COVERAGE:
--------------------------------------------------------------------------------------------------
Name                                    Design Design   Lang File(Line)      Hits Status
                                        Unit   UnitType
--------------------------------------------------------------------------------------------------
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ack    FIFO_sva Verilog  SVA  FIFO_sva.sv(129)1277 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ack_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(130)1463 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow  FIFO_sva Verilog  SVA  FIFO_sva.sv(131) 263 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(132)2477 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow FIFO_sva Verilog  SVA  FIFO_sva.sv(133) 415 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(134)2325 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_full      FIFO_sva Verilog  SVA  FIFO_sva.sv(135) 399 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_empty     FIFO_sva Verilog  SVA  FIFO_sva.sv(136) 755 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_almostfull
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(137) 309 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_almostempty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(138) 529 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr  FIFO_sva Verilog  SVA  FIFO_sva.sv(139) 778 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_rd  FIFO_sva Verilog  SVA  FIFO_sva.sv(140) 338 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_idle
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(141) 610 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_empty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(142) 163 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_full
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(143)  88 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_mid
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(144) 336 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_inc
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(145)1277 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_inc
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(146) 762 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_wrap
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(147) 109 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_wrap
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(148)  53 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(149)1463 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(150)1978 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_data_out_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(151)1978 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_fill_to_full
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(153)  64 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_drain_to_empty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(154)   5 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow_burst
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(155) 121 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow_burst
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(156) 170 Covered
```

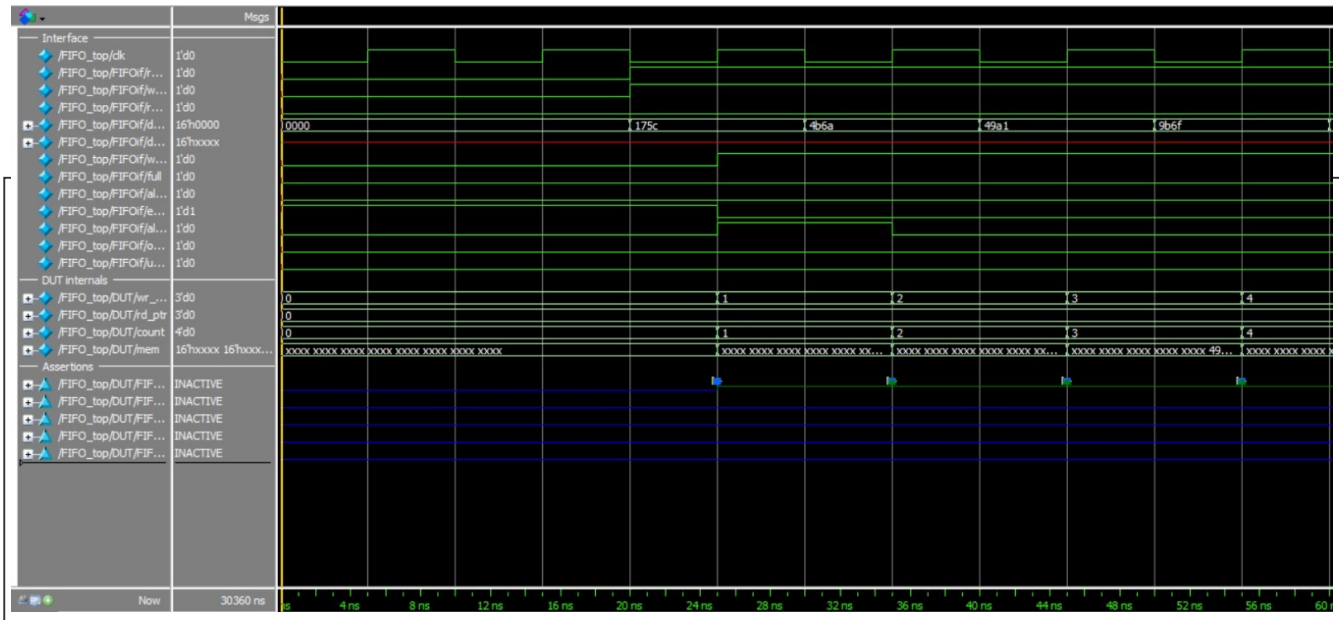
```
DIRECTIVE COVERAGE:
--------------------------------------------------------------------------------------------
Name                                    Design Design   Lang File(Line)      Hits Status
                                        Unit   UnitType
--------------------------------------------------------------------------------------------
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ack    FIFO_sva Verilog  SVA  FIFO_sva.sv(129)1277 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ack_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(130)1463 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow  FIFO_sva Verilog  SVA  FIFO_sva.sv(131) 263 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(132)2477 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow FIFO_sva Verilog  SVA  FIFO_sva.sv(133) 415 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow_low
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(134)2325 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_full      FIFO_sva Verilog  SVA  FIFO_sva.sv(135) 399 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_empty     FIFO_sva Verilog  SVA  FIFO_sva.sv(136) 755 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_almostfull
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(137) 309 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_almostempty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(138) 529 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr  FIFO_sva Verilog  SVA  FIFO_sva.sv(139) 778 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_rd  FIFO_sva Verilog  SVA  FIFO_sva.sv(140) 338 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_idle
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(141) 610 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_empty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(142) 163 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_full
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(143)  88 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_count_wr_rd_mid
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(144) 336 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_inc
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(145)1277 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_inc
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(146) 762 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_wrap
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(147) 109 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_wrap
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(148)  53 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_wr_ptr_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(149)1463 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_rd_ptr_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(150)1978 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_data_out_hold
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(151)1978 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_fill_to_full
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(153)  64 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_drain_to_empty
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(154)   5 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_overflow_burst
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(155) 121 Covered
/FIFO_top/DUT/FIFO_sva_inst/cp_underflow_burst
                                        FIFO_sva Verilog  SVA  FIFO_sva.sv(156) 170 Covered

TOTAL DIRECTIVE COVERAGE: 100.00%  COVERS: 27

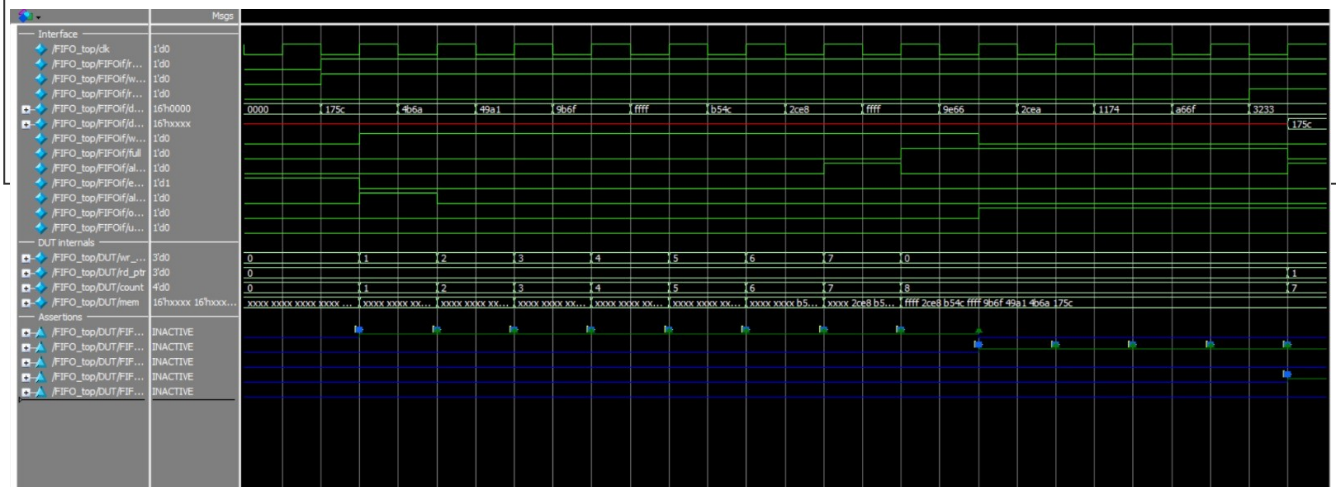
Total Coverage By Instance (filtered view): 100.00%
```

Waveforms:

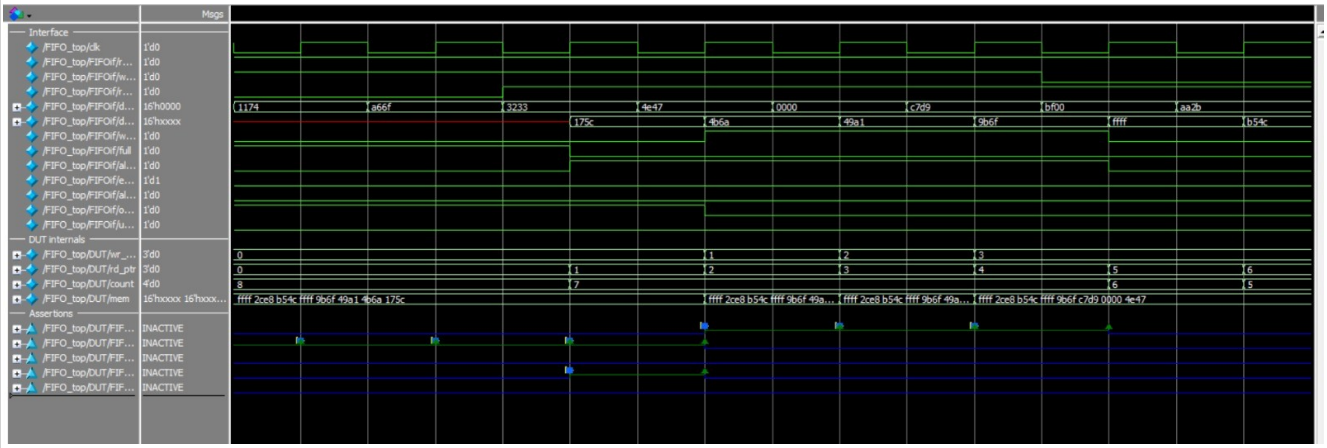
FIFO_reset_sequence:



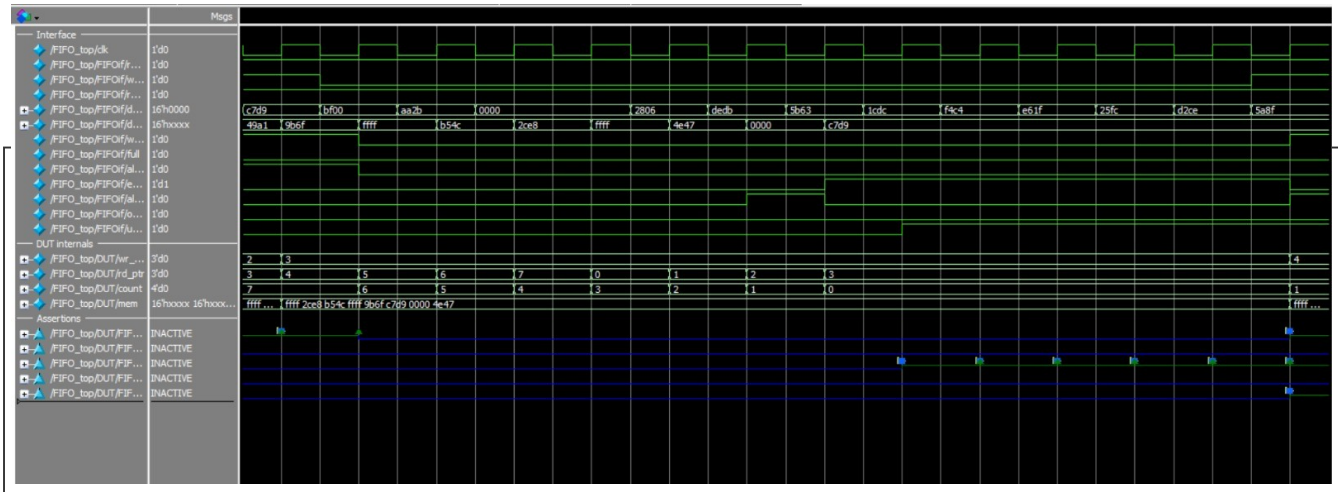
FIFO_write_only_sequence:



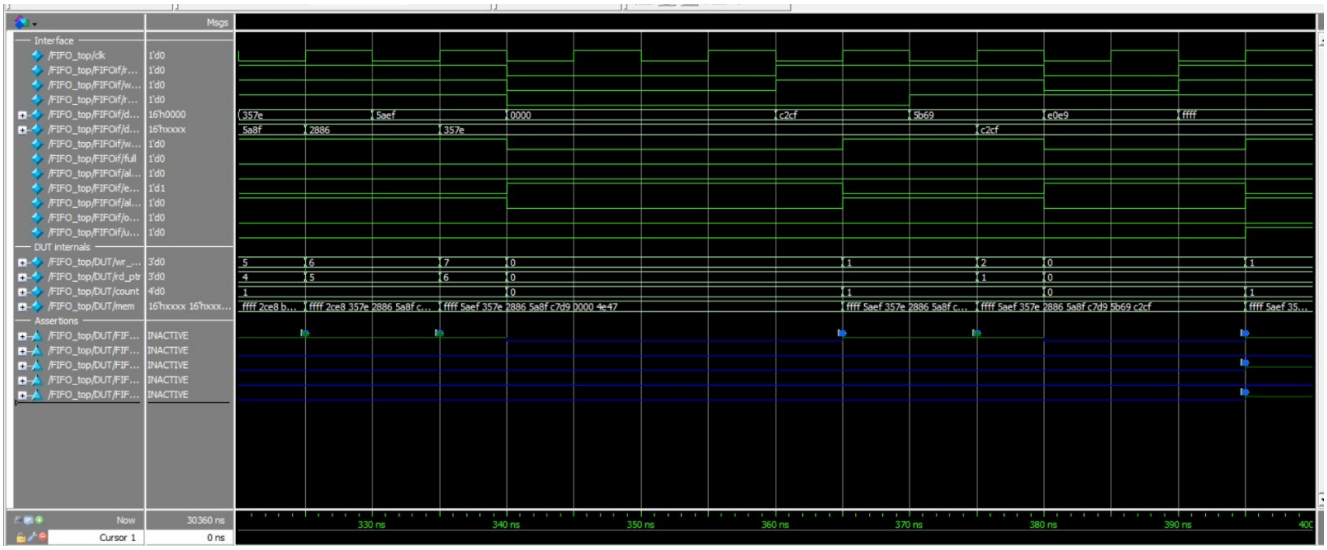
FIFO_write_read_sequence:



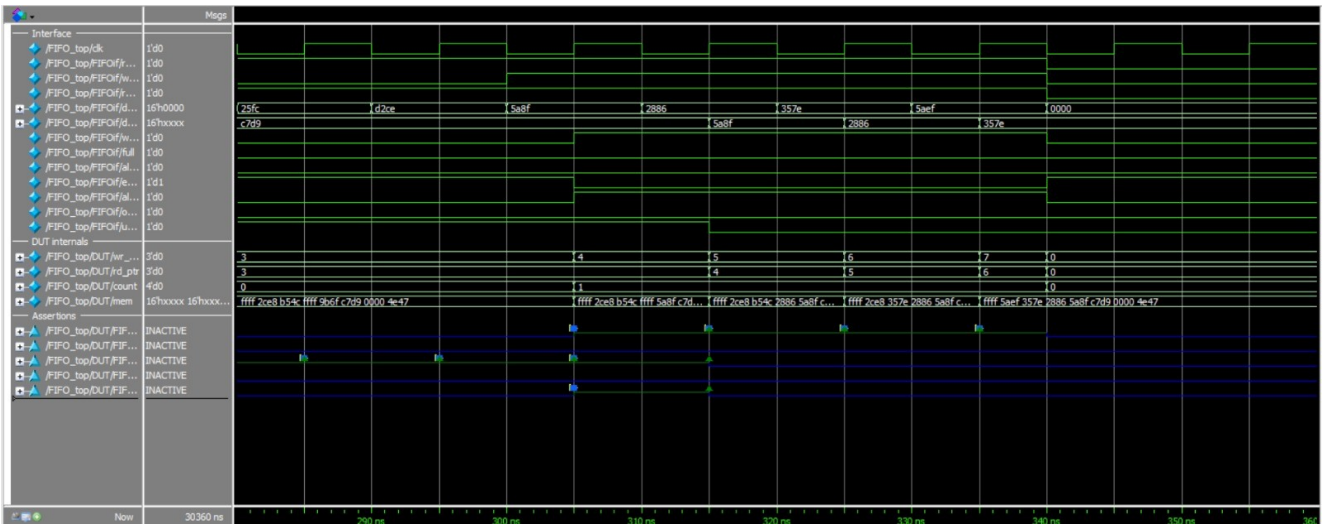
FIFO_read_only_sequence:



FIFO_reset_sequence:



FIFO_write_read_sequence :



FIFO_main_sequence:

